Matched regularization of first- and second-block $d4$ routes

The adapter control restores an accessibility-selected compositional regime

Design and validation

This experiment protects the pretrained $d2$ parents while exposing newly learned $d4$ routes in both blocks to the same regularization. After parent pretraining, E and the rescaled base V_1 are frozen. A small parallel trainable adapter V_1' is added, and phase 2 applies the same coefficient λ to V_1', V_2, and the $d4$ readout row W_4. The two $d2$ losses remain active. As in the earlier accessibility sweeps, $V_1 \mapsto \alpha V_1$ controls the measured onset ratio $R_0 = Q_2/Q_1$; the adapter starts at 10^{-3} of standard initialization and negligibly perturbs R_0.

The completed grid contains eight α values, eight decay values, and three pretrained seeds. All 192 result cells and all 192 trajectory cells are present. Unless stated otherwise, numbers and phase colors below are three-seed means. The red line marks the theoretical frozen-feature threshold $R_c = 1/\sqrt{8} = 0.354$.

Main result: a compositional phase returns, but not exactly at R_c

Matched adapter decay produces a clear blue-to-red boundary that was absent from the matched-V_2/W_4 control. At $\lambda = 10^{-2}$, the mean block-2 2+2 fraction rises from 0.010 ± 0.004 at $R_0 = 0.323$ to 0.340 ± 0.119 at $R_0 = 0.416$, then to 0.880 ± 0.015 at $R_0 = 0.515$ and 0.899 ± 0.019 at $R_0 = 0.573$. The same structure is already visible at $\lambda = 3 \cdot 10^{-3}$ (0.028, 0.318, 0.848, 0.888 at those four R_0 values). Thus the theoretical threshold correctly identifies the side on which composition becomes possible, but the empirical transition is broad and displaced toward larger accessibility. Its seed sensitivity is localized near the boundary: the largest standard deviation is 0.119 at $(\alpha, \lambda) = (1.5, 10^{-2})$, whereas the high-$R_0$ phase has standard deviation at most 0.020 for $\lambda \geq 3 \cdot 10^{-3}$.

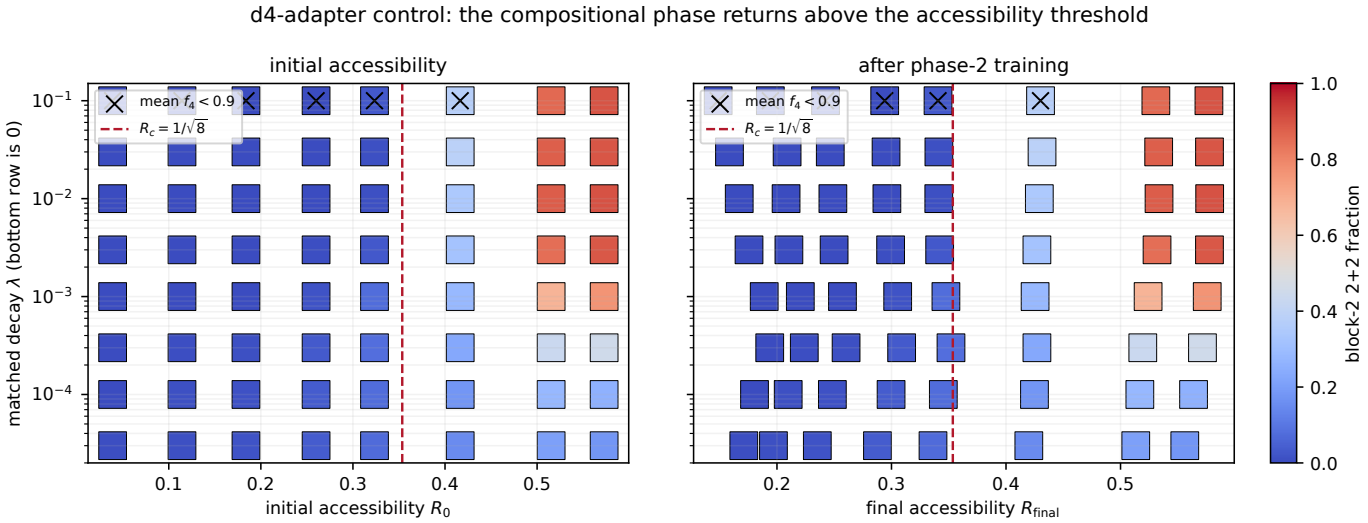


Figure 1: Mean block-2 2+2 fraction indexed by initial (left) and final (right) accessibility. Crosses mark cells with mean $d4$ coefficient below 0.9, which occurs mainly at the strongest decay and should not be read as a clean circuit-selection result.

Comparison with the earlier controls

The adapter intervention reverses the failure of the fully trainable controls. At $R_0 = 0.416, \lambda = 10^{-2}$, the 2+2 fraction is 0.340 with the adapter, versus 0.159 under V_2-only decay and 0.0007 under matched V_2/W_4 decay. At $R_0 = 0.515$ the comparison is 0.880 versus 0.230 and 0.0002. The frozen-parent benchmark remains sharper: at nearby $R = 0.438$ and 0.499, it reaches 0.904 and 0.975. The adapter therefore recovers most of the frozen compositional solution at high accessibility, while the onset cell remains mixed and variable.

This result identifies the missing condition in the earlier experiments. Matching the V_2 and readout penalties alone closes a scale escape but leaves the base representation free to reorganize and bypass parent composition. Freezing the pretrained parent-producing map prevents that reorganization. Under this constraint, matched regularization again makes initial accessibility predictive of the block-2 circuit.

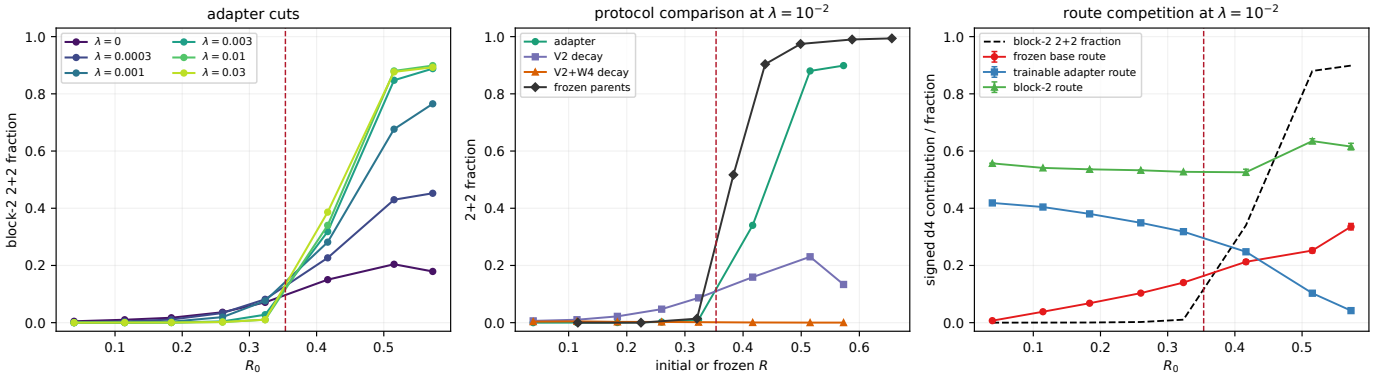


Figure 2: Left: fixed-λ cuts through the adapter phase diagram. Center: direct comparison at $\lambda = 10^{-2}$. Right: signed $d4$ readout contributions from the frozen base, trainable adapter, and block 2; the dashed line is the internal block-2 2+2 fraction. Error bars are one standard deviation across seeds.

Route competition: block 2 composes, while block 1 remains direct

The adapter's classified computation is 1+1+1+1 in every cell ($f_{2+2}^{\text{adapter}} < 1.3 \cdot 10^{-33}$). It is therefore the intended direct first-block competitor, not another route to parent composition. At $\lambda = 10^{-2}$ and $R_0 = 0.416$, the mean signed contributions to $\hat{f}_4 = 0.986$ are 0.213 from the frozen base, 0.248 from the adapter, and 0.526 from block 2. At $R_0 = 0.515$ they are 0.252, 0.103, and 0.635; at $R_0 = 0.573$, 0.336, 0.042, and 0.616.

Thus increasing accessibility suppresses the penalized direct adapter and selects a majority block-2 route whose internal computation is 88–90% compositional. It does *not* produce a pure end-to-end compositional phase: the unpenalized frozen base still carries 25–34% of $d4$ in the high-R_0 cells. This fixed residual shortcut is now the main obstruction, rather than a trainable first block learning around the penalty.

Accessibility drift and norm balance

$R(t)$ changes much less than in the earlier fully trainable sweeps. At $\lambda = 10^{-2}$, $R_0 = 0.323, 0.416, 0.515, 0.573$ end at $R_{\text{final}} = 0.341, 0.429, 0.534, 0.578$. Very low-$R_0$ cells rise more ($0.039 \to 0.167$), but no trajectory crosses from the clearly subcritical group into the high-composition phase. The circuit change is consequently associated with the controlled initial representation rather than a common late-time attractor in R.

The matched penalty also avoids the catastrophic scale imbalance of V_2-only decay. At $\lambda = 10^{-2}$, the norms (V_1', V_2, W_4) are $(1.041, 1.200, 1.643)$ for $R_0 = 0.416$, $(0.543, 1.124, 1.404)$ for $R_0 = 0.515$, and $(0.266, 0.901, 1.145)$ for $R_0 = 0.573$. In particular, $\|W_4\|/\|V_2\| = 1.25$–1.37 in the transition and compositional regime, compared with 3.22 at $R_0 = 0.416$ in the matched-V_2/W_4 experiment and an essentially collapsed V_2 under V_2-only decay. The frozen benchmark is more nearly balanced still (ratio about 1.08 at nearby R), but the adapter control eliminates the qualitative scale escape.

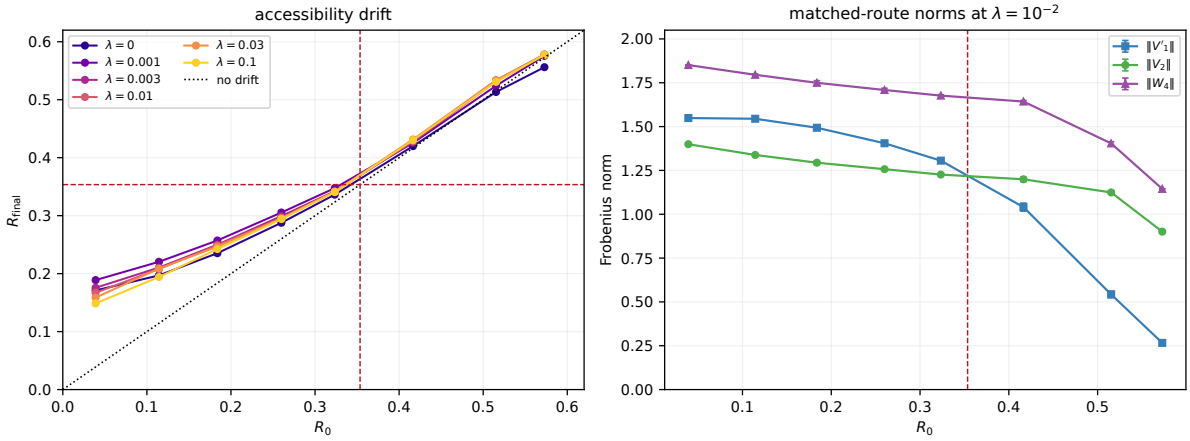


Figure 3: Left: R_{final} versus R_0. Right: the three matched-route norms at $\lambda = 10^{-2}$. The direct adapter norm falls sharply as the compositional block-2 route becomes accessible.

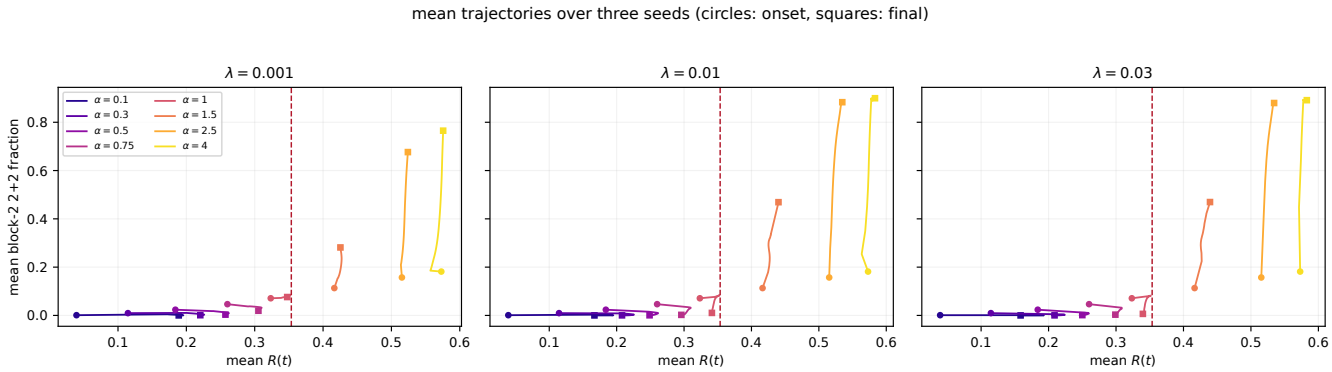


Figure 4: Mean trajectories over the three seeds. Above the accessibility boundary, block-2 composition grows rapidly without a large accompanying drift in $R(t)$; below it, the trajectories remain direct.

Conclusion. Freezing the pretrained $d2$ parents and taxing the new direct and compositional routes on equal footing restores the qualitative accessibility transition predicted by the frozen theory. The transition is not pinned sharply to $1/\sqrt{8}$: it occupies the interval $R_0 \simeq 0.42$–0.52 and is most variable at its lower edge. High accessibility selects a robust, majority block-2 route with nearly pure 2+2 internal structure. A pure network-level compositional phase is still prevented by the frozen first-block $d4$ shortcut, whose readout contribution remains substantial precisely where the adapter route has been regularized away.

Artifacts are stored in the d4-adapter analysis directory: merged results and trajectories, phase diagrams, cuts, protocol comparison, route decomposition, drift, norm balance, and phase-plane trajectories. Training used `analysis/accessibility/run_alpha_unfrozen_experiment.py` with `--wd-d4-adapter`.